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|----|----------------------------------------------------------------------------------------|----------|
| 04 | <i>Optimization of Bivariate functions and its applications for Business students.</i> | A2,Q2,M2 |
| 05 | <i>Integral calculus and area problems in Economics</i>                                | A3, Q3,F |

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|--------------|--------|----------------------------------------------------------------------------------------------------------|
| Text Book(s) | Title  | <i>Applied Mathematics for Business, Economics, and Social Sciences</i> , 4 <sup>th</sup> Ed., Singapore |
|              | Author | Frank S. Budnick                                                                                         |
| Ref. Book(s) | Title  | <i>Essential Mathematics for Economics and Business</i> , 3 <sup>rd</sup> Ed.                            |
|              | Author | Teresa Bradley                                                                                           |

| Week | Course Contents                                                                                                                                                                                    | Chapter | CLO |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----|
| 01   | Brief Synopsis of the Course, Quadratic function, Exponential function, Logarithmic function<br>Applications: supply, demand, revenue, cost                                                        | 6,7     | 1   |
| 02   | Introduction, Graphical Solutions (The Graphics of Linear Inequalities, system of linear Inequalities, Region Of Feasible Solutions, Incorporating the Objective Function, Corner- point Solutions | 10      | 2   |
| 03   | Alternative Optimal Solutions,<br>No Feasible Solution, Unbounded Solution.<br>The Simplex Method (Maximization and Minimization Problems)                                                         | 10+11   | 2   |
| 04   | Application of Linear Programming, Phenomena of alternative optimal solutions, no feasible solution, Unbounded solution.                                                                           | 11      | 2   |
| 05   | <b>MID-TERM-I</b>                                                                                                                                                                                  |         |     |
| 06   | Depreciation, Annuities and their future value                                                                                                                                                     | 8       | 3   |
| 07   | Annuities and their present value                                                                                                                                                                  | 8       | 3   |
| 08   | Partial Derivatives, Optimization of Bivariate Functions                                                                                                                                           | 20      | 4   |
| 09   | Application of Bivariate Optimization                                                                                                                                                              | 20      | 4   |
| 10   | Integration of transcendental functions<br>Methods of integration                                                                                                                                  | 18      | 5   |
| 11   | <b>MID-TERM-II</b>                                                                                                                                                                                 |         |     |
| 12   | The definite integral and area under the curve                                                                                                                                                     | 19      | 5   |





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|---------------------------|----------------------------------------------------------------------|--------------------|-----------------|
| <b>Department</b>         | Fast School of Management                                            | <b>Dept. Code</b>  | BBA, BSAF, BSBA |
| <b>Course Title</b>       | Business Mathematics 2                                               | <b>Course Code</b> | MT 2004         |
| <b>Pre-requisite(s)</b>   | Business Mathematics 1                                               | <b>Credit Hrs.</b> | 3               |
| <b>Moderator</b>          | Dr. Hina Firdous                                                     |                    |                 |
| <b>Course Instructors</b> | Dr. Hina Firdous, Dr. Qasim Noor, Mr. Saleem Chishti, Ms Iqra Yaqoot |                    |                 |
| <b>Note:</b>              | It is a tentative schedule of course. It may vary (if required).     |                    |                 |

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| <b>Course Objective</b> | <p>Business Mathematics presents Mathematical skills and knowledge that you can apply to solve financial problems. Finance plays a major role in our lives e.g. car loans, mortgage payments, retirement plans, real estate investment and knowing how to calculate the cost of borrowing or the return on investment is important to us. This course provides step-by-step guidance through sample problems and solutions related to banking, credit, basic finance and investment. Through this course students can gain an understanding of financial instruments and terminology used in business finance such as compound interest, annuities and promissory notes. This course also gives knowledge about how to use calculus in the field of business and management. The course serves as a good foundation for further study in management, accounting and finance.</p> |
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*I = Introduction, R = Reinforcement, E = Evaluation.*

*A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.*

| No. | Course Learning Outcome (CLO) Statements                                                          | Tools    |
|-----|---------------------------------------------------------------------------------------------------|----------|
| 01  | <i>To understand the basic Non Linear function and its applications in Business and Economics</i> | A1,Q1    |
| 02  | <i>To learn Linear Programming and its importance</i>                                             | A1,Q1,M1 |
| 03  | <i>Financial Mathematics Annuities, Debt Repayments, Sinking funds</i>                            | A2, Q2   |



|            |                                                               |    |   |
|------------|---------------------------------------------------------------|----|---|
| 12         | Methods of Approximation<br>Rectangle Rule, Trapezoidal Rule, | 19 | 5 |
| 13         | Simpson's Rule                                                | 19 | 5 |
| 14         | Application of Integral Calculus                              |    |   |
| 15         | Differential Equations                                        | 18 | 5 |
| FINAL EXAM |                                                               |    |   |

**Evaluation Procedure & Marks Distribution:**

| Assessment Tools | Total No. of | Weightage |
|------------------|--------------|-----------|
| Quizzes          | Minimum 3    | 10%       |
| Assignments      | Minimum 3    | 10%       |
| Mid Term Exam    | 2 (I+II)     | 30%       |
| Final Exam       | 1            | 50%       |

**Note:** No homework/ assignment submission after due date. Relative grading policy will be followed.